# THE UNIVERSITY OF HONG KONG
# FACULTY OF BUSINESS AND ECONOMICS

## MFIN7068 Crypto Finance

**GENERAL INFORMATION**

Instructor: Gracy Chen (CEO of Bitget)
Email: TBD
Tutor: TBD
Consultation times: TBD

Course Website: Moodle / TBD
Reference Book: The course will primarily draw upon lecture slides, readings and course-provided materials.

**COURSE DESCRIPTION**

This course explores the latest trends in crypto finance (2025), covering Bitcoin, Ethereum, DeFi, stablecoins, RWAs, AI-driven protocols, institutional adoption, DATs, and regulatory shifts. Designed for finance and mathematics students, it blends case studies, real-world applications, and strategic insights from Gracy Chen and top industry leaders.

Rather than pure theory, the course focuses on actionable knowledge—how crypto finance operates today, the mistakes of past cycles, and emerging opportunities in institutional crypto, DeFi 3.0, and Web3 venture building.

**COURSE OBJECTIVES**

1. Develop a comprehensive understanding of advanced crypto trends. This involves Bitcoin, Ethereum, Stablecoins, DEFi, RWAs, AI-driven protocols, DATs, regulatory shifts and institutional adoption.
2. Obtain practical experience building and investing in the crypto space.

**PROGRAMME LEARNING OUTCOMES**

PLO1. Acquisition of the techniques and advanced knowledge in the interdisciplinary fields of financial technology including but not limited to financial services, technology and law. Students should acquire techniques and advanced knowledge in the interdisciplinary fields of financial technology, including but not limited to financial services, technology and law.

PLO2. Application and integration of interdisciplinary knowledge and skills to identify and tackle practical problems and design innovative products and systems with international standards and global vision. Students should be able to apply and integrate interdisciplinary knowledge and skills to identify and tackle problems and design innovative products and systems with international standards and global vision. They should have an appreciation for diversity and cultural differences and be able to function in multi-cultural settings.

PLO3. Inculcating leadership, professional ethics and competence in the interdisciplinary fields of financial technology. Students should demonstrate the capacity to recognise, understand and develop appropriate options in ethical and social issues in the interdisciplinary fields of financial technology. In their learning experiences, they will be working in teams that promote leadership training and development of leadership traits. In addition, they are to adhere to and respect university regulations on academic conduct.

PLO4. Mastering communication skills. Students should be able to articulate in public and be able to make convincing and coherent presentations, write effectively and use specific technical terminology in contemporary presentation tools context.

**COURSE LEARNING OUTCOMES**

| Course Learning Outcomes | Aligned Programme Learning Outcomes |
|---|---|
| CLO1: Develop a comprehensive understanding of advanced techniques in the crypto space, such as technical analysis, chain analysis, and fundamental analysis of crypto projects | PLO1, 2 |
| CLO2: Apply advanced analytical methods, including technical, chain, and fundamental analysis, to assess the value and potential of current and emerging crypto projects | PLO1, 2 |
| CLO3: Assess the impact of emerging trends—such as DeFi, stablecoins, RWAs, AI-driven protocols, and institutional adoption—on contemporary crypto finance | PLO1, 2, 3 |
| CLO4: Analyse previous crypto market cycles and identify actionable opportunities for institutional crypto, DeFi 3.0, and Web3 venture building | PLO1, 2, 3 |
| CLO5: Create and deliver professional investment presentations and concise decks tailored to the expectations of crypto investors and industry practitioners | PLO3, 4 |

**COURSE TEACHING AND LEARNING ACTIVITIES**

| Course Teaching and Learning Activities | Expected Hours | Study Load (% of study) |
|---|---|---|
| T&L1. Interactive lectures | 30 | 25% |
| T&L2. Assignments | 60 | 50% |
| T&L3. Lecture preparation & Self-study | 30 | 25% |
| Total | 120 | 100% |

| Assessment Methods | Brief Description (Optional) | Weight | Aligned Course Learning Outcomes |
|---|---|---|---|
| A1. In-class & forum participation & Quiz | Attendance & participation | 20% | CLO1, 2, 4 |
| A2. Assignment | Effort and accuracy | 30% | CLO1, 2, 3, 4 |
| A3. Web3 Protocol Exploration Lab | Continuous assessment where students apply concepts | 10% | CLO1, 2 |
| A4. Final Project | Culminating presentation | 40% | CLO1, 2, 3, 4, 5 |
| | Total | 100% | |

| STANDARDS FOR ASSESSMENT | |
|---|---|
| **Course Grade Descriptors** | |
| A+, A, A- | •Demonstrate a strong understanding of all relevant knowledge<br>•Handling questions professionally<br>•High participation in discussions<br>•Present arguments that have an element of originality<br>•Excellent performance in assignments |
| B+, B, B- | •Demonstrate a good understanding of all relevant knowledge<br>•Handling questions in a logical way<br>•Good participation in discussions<br>•Present arguments that go beyond the lecture and textbook<br>•Good performance in assignments |
| C+, C, C- | •Demonstrate a basic understanding of the concepts involved<br>•Fairly address questions as set<br>•Some participation in discussions<br>•Present arguments in a well-structure manner<br>•Acceptable performance in assignments |
| D+, D | •Demonstrate a minimum understanding of the concepts involved<br>•Barely address questions as set<br>•Minimal or no participation in discussions<br>•Present arguments in a marginally acceptable manner<br>•Marginally acceptable performance in assignments |
| F | •Demonstrate a poor understanding of the concepts involved<br>•Unable or unwilling to handle questions<br>•Minimal or no participation in discussions<br>•Present arguments poorly<br>•Poorly performance in assignments |
| **Assessment Rubrics for Participation** | |
| A+, A, A- | •High participation in discussions<br>•Always attend in-class discussions<br>•Demonstrate a strong understanding of all relevant knowledge<br>•Handling questions professionally<br>•Present arguments that have an element of originality<br>•Respect others and follow the class rules (no chatting and do not use cell phone) |
| B+, B, B- | •Good participation in discussions<br>•Often attend in-class discussions<br>•Demonstrate a good understanding of all relevant knowledge<br>•Handling questions in a logical way<br>•Present arguments that go beyond the lecture and textbook<br>•Respect others and follow the class rules (no chatting and do not use cell phone) |
| C+, C, C- | •Some participation in discussions<br>•Sometimes attend in-class discussions<br>•Demonstrate a basic understanding of the concepts involved<br>•Fairly address questions as set<br>•Present arguments in a well-structure manner<br>•Respect others and follow the class rules (no chatting and do not use cell phone) |
| D+, D | •Minimal or no participation in discussions<br>•Rarely attend in-class discussions<br>•Demonstrate a minimum understanding of the concepts involved<br>•Barely address questions as set<br>•Present arguments in a marginally acceptable manner<br>•Respect others and follow the class rules (no chatting and do not use cell phone) |
| F | •Minimal or no participation in discussions<br>•Almost never attend in-class discussions<br>•Demonstrate a poor understanding of the concepts involved |

| | |
|---|---|
| | •Unable or unwilling to handle questions<br>•Present arguments poorly<br>•Behave poorly in class (often chatting with others, using cell phones, or being late) |
| **Assessment Rubrics for Assignments and the Exam** | |
| A+, A, A- | •Demonstrate a strong understanding of all relevant knowledge<br>•Present arguments that have an element of originality<br>•Achieve a standard of excellent performance in the assessments with very accurate computation and very good analytical and problem solving skills |
| B+, B, B- | •Demonstrate a good understanding of all relevant knowledge<br>•Present arguments that go beyond the lecture and textbook<br>•Achieve a standard of good performance in the assessments with accurate computation and good analytical and problem solving skills |
| C+, C, C- | •Demonstrate a basic understanding of the concepts involved<br>•Present arguments in a well-structure manner<br>•Meet a standard of acceptable performance in the assessments with reasonably accurate computation and acceptable analytical and problem solving skills |
| D+, D | •Demonstrate a minimum understanding of the concepts involved<br>•Present arguments in a marginally acceptable manner<br>•Meet a standard of marginally acceptable performance in the assessments with some errors in computation and barely adequate analytical and problem solving skills |
| F | •Demonstrate a poor understanding of the concepts involved<br>•Present arguments poorly<br>•Fail to meet a standard of passing the assessments with major errors in computation and inadequate analytical and problem solving skills |

**COURSE CONTENT AND TENTATIVE TEACHING SCHEDULE**

**Subject to change (indicative list)**

1. Week 1 – Blockchain, Bitcoin and Ethereum

Money → digital; PoW/PoS; Bitcoin whitepaper; ETFs, L2s, restaking, DATs.

Week 2 – Markets, Trading, RWAs, Institutions

Exchange stack, derivatives, on-chain/AI bots, tokenised treasuries, stablecoins, ETF flows, DAT case studies.

Week 3 – Global Regulation & Risk

MiCA live, US post-election, HK/SG/UAE 2025 rules, El Salvador BTC legal-tender case, 3AC case, Risk Management

Week 4 – DeFi, GameFi, SocialFi, NFT, DePIN

DEX/lending, yield farming, DAOs, Web3 gaming, SocialFi, NFT use-cases, decentralised infra networks; hands-on labs.

Week 5 – Venture Building & Emerging Tech

Tokenomics, regulatory-safe fundraising, VC & secondary desks, scaling, MPC vs multisig, AI+ZK trends.

2. Students may also expect guest speakers, subject to avaiability. In the past, this has meant fireside chats and live Q&A with founders of restaking and move-to-earn protocols, CEOs of Ethereum-infrastructure and Web3-gaming giants, partners at tier1 crypto VCs, senior legal officers from sovereign Bitcoin offices and HK regulators, plus ex-CTOs who have taken high-frequency exchanges from zero to production.

**REQUIRED/RECOMMENDED READINGS & ONLINE MATERIALS** (e.g. journals, textbooks, website addresses etc.)

Course materials will be provided. The professor will provide various readings.

Here are some sample readings from the past.

1. Bitcoin whitepaper – https://bitcoin.org/bitcoin.pdf
2. "How does a blockchain work" video – https://www.youtube.com/watch?v=SSo_EIwHSd4
3. ConsenSys – https://consensys.io/
4. HK stablecoins article – https://www.regulationtomorrow.com/asia/hong-kong-stablecoins-sprint-historic-bill-passes-draft-licensee-guidelines-released-and-aml-cft-consultation-begins/
5. Animoca RWA report – https://docsend.com/view/m7n5bkjj98ud2kvt
6. MicroStrategy HBS case – https://store.hbr.org/product/microstrategy-s-investment-in-bitcoin/222048
7. El Salvador HBS case – https://www.hbs.edu/faculty/Pages/item.aspx?num=62068
8. STEPN Ivey case – https://store.hbr.org/product/stepn-preempting-a-death-spiral/W34582
9. 2025 DePIN report – https://www.tbstat.com/wp/uploads/2025/07/DePIN_Report_Final.pdf
10. Three Arrows Capital HBS case – https://store.hbr.org/product/three-arrows-capital-a-crypto-hedge-fund-failure-and-operational-due-diligence-lessons/HK1439
11. samczsun blogs – https://www.paradigm.xyz/team/samczsun

**MEANS/PROCESSES FOR STUDENT FEEDBACK ON COURSE**

- o Online response via Moodle site
- ✓ Others: ___SFTL_____ (please specify)

**COURSE POLICY** (e.g. plagiarism, academic honesty, attendance, etc.)

An orderly learning environment is extremely important for this course. Disruptive behaviors are absolutely unacceptable. Academic dishonesty includes cheating, plagiarism, unauthorized collaboration, falsifying academic records, and any act designed to avoid participating honestly in the learning process. Any such dishonesty will result in an F grade.

**ADDITIONAL COURSE INFORMATION** (e.g. e-learning platforms & materials, penalty for late assignments, etc.)

Assignments will be submitted in class before the lectures. No late assignment will be accepted.
Lecture notes and self-learning materials will be uploaded on Moodle.
Good questions and discussions on Moodle will be awarded participation grades.